Mixture Model Challenge

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March 27, 2026

Recapitulation

- What is latent variable of mixture model?

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- What is the conditional likelihood of observation of a mixture model?

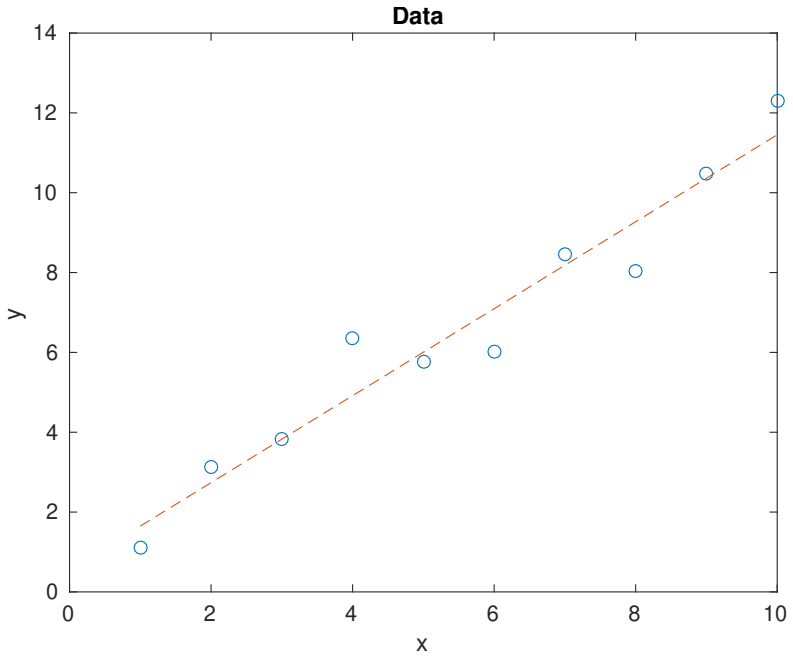
Recapitulation

- What is latent variable of mixture model?
- What is the conditional likelihood of observation of a mixture model?
- What is the advantage of one-hot representation?

Review of building blocks

Linear regression:

$$
\begin{aligned}
y_i &= \boldsymbol{x}_i\theta + e_i, \\
e_i &\sim \mathcal{N}(0, \sigma_e), \\
p(y_i|x_i, \theta) &= \mathcal{N}(\boldsymbol{x}_i\theta, \sigma_e),
\end{aligned}
$$



Review of building blocks

Linear regression likelihood:

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\end{aligned}
$$

Prior (ridge):

$$p(\theta_i) = \mathcal{N}(0, \alpha)$$

Sparsity:

$$
\begin{aligned}
p(\theta_i) &= St(\nu, \theta, \sigma_0) \\
&= \int \mathcal{N}(0, \alpha_i) p(\alpha_i) d\alpha_i \\
p(\theta_i) &= 0.9\mathcal{N}(0, \sigma_0) + 0.1\mathcal{N}(0, 100\sigma_0) \\
&= \sum_{l=1}^{2} \mathcal{N}(0, \sigma_l) p(l)
\end{aligned}
$$

Review of building blocks

Linear regression likelihood:

$$y_i = \boldsymbol{x}_i\theta + e_i,$$
$$e_i \sim \mathcal{N}(0, \sigma_e),$$
$$p(y_i|x_i, \theta) = \mathcal{N}(\boldsymbol{x}_i\theta, \sigma_e),$$

Outliers:

$$p(y_i|\boldsymbol{x}, \theta) = St(\nu, \boldsymbol{x}_i\theta, \sigma_0)$$
$$p(y_i|\boldsymbol{x}, \theta) = 0.9\mathcal{N}(\boldsymbol{x}_i\theta, \sigma_0) + 0.1\mathcal{N}(\boldsymbol{x}_i\theta, 100\sigma_0)$$

Prior (ridge):

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Sparsity:

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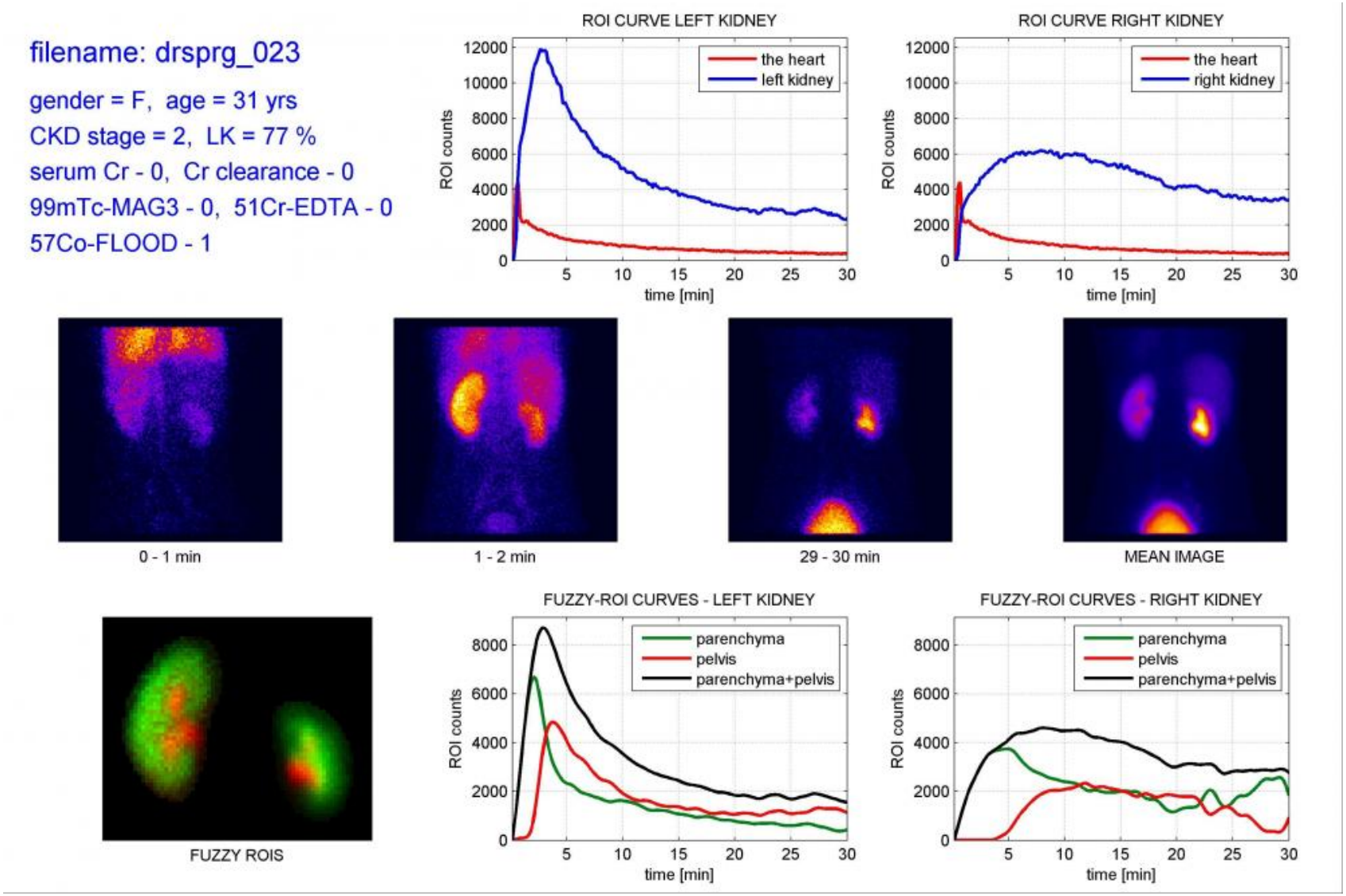
$$= \sum_{l=1}^{2} \mathcal{N}(0, \sigma_l)p(l)$$

Patlak Rutland plot

Sequence of scintigraphic images of kidneys.

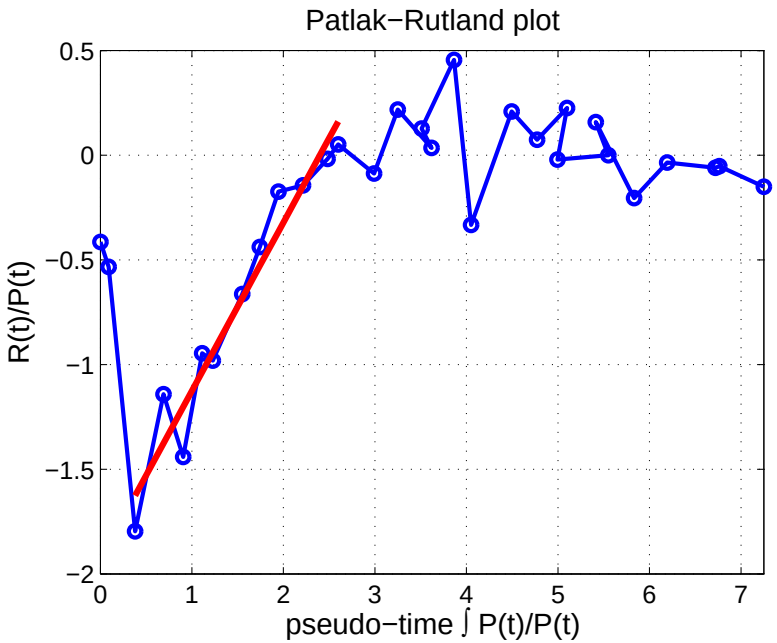


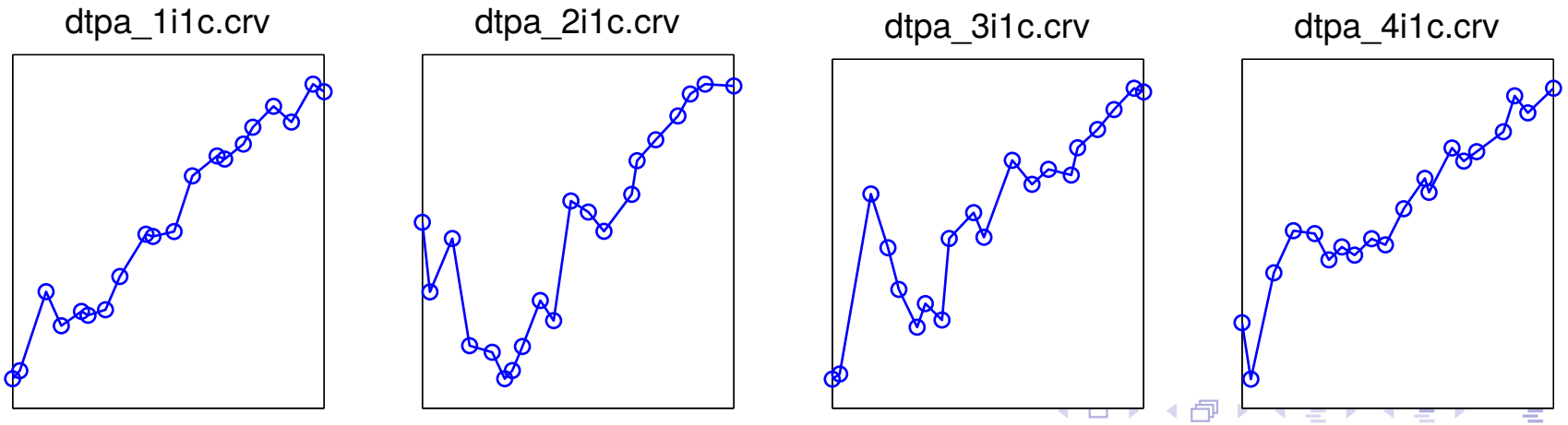
Challenge: Patlak Ruland plot

Patlak Rutland plot is a ratio of the parenchyma curve over the cumulative sum of the heart curve.

Diagnostically important is the **slope** of the linear part, which:

- typically starts around 1min
- typically ends around 3min
- with outliers
- manual fit is the norm





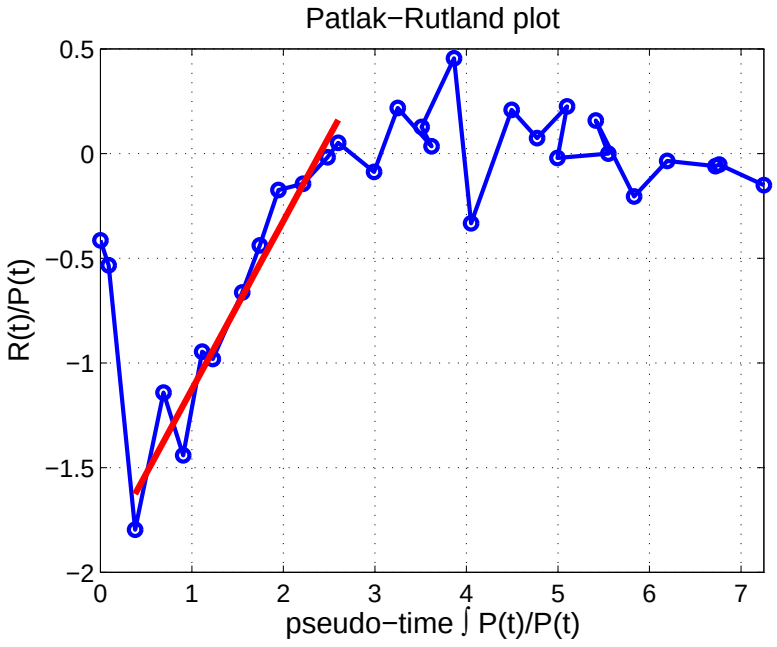
Models?

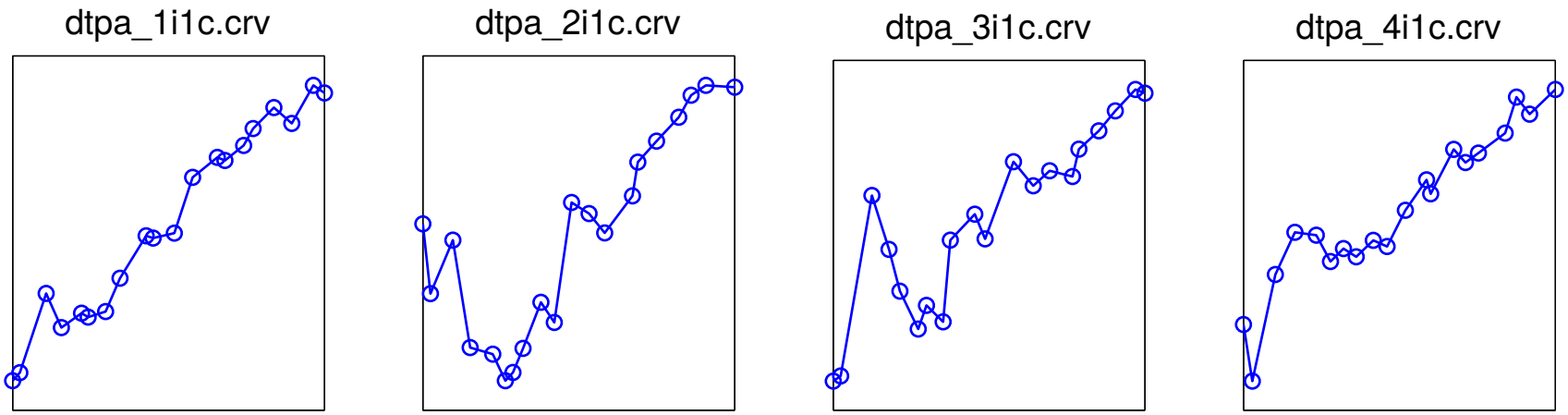
- no heuristics
- probabilistic model providing either:
  - estimate of $\hat{a}$
  - posterior distribution $p(a)$

  where a is the slope of:

$$y_i = ax_i + b + e_i$$

  for points belonging to it.



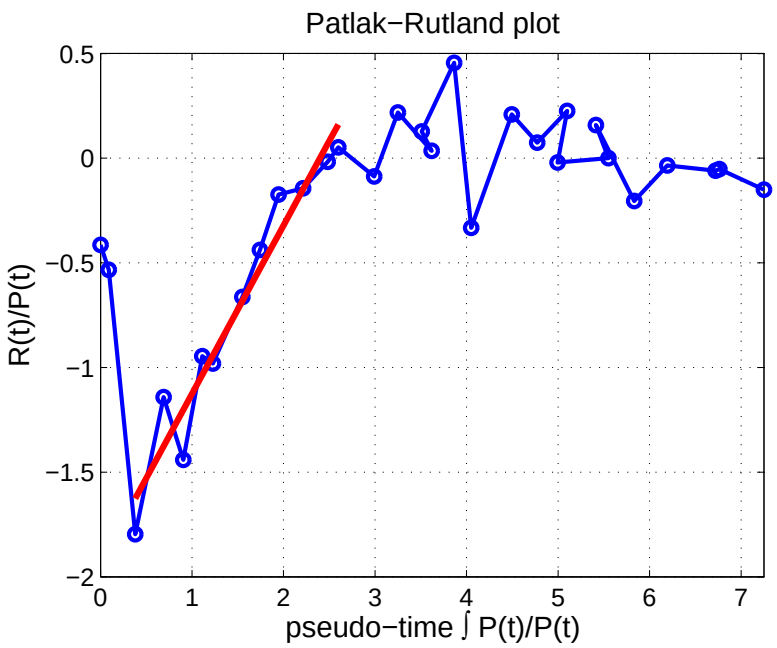


Model 1: Robust regression

Model

$$y_i = ax_i + b + e_i$$

is valid for all $x_i \in [1, 3]$ with heavy-tailed noise $p(e_i)$.



Model 1: Robust regression

Model

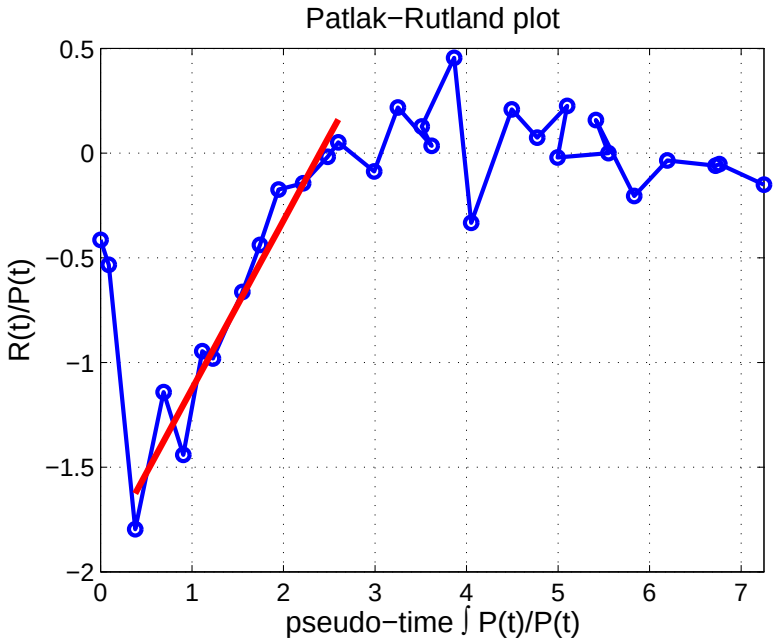
$$y_i = ax_i + b + e_i$$

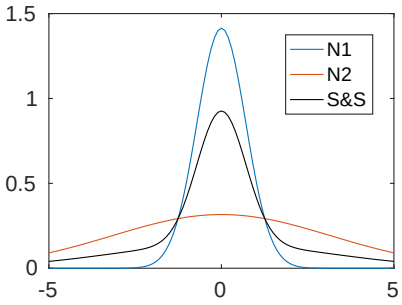
is valid for all $x_i \in [1, 3]$ with heavy-tailed noise $p(e_i)$.
Possibilities:

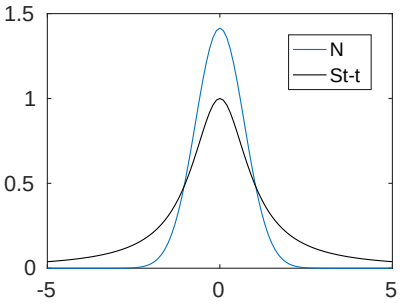
- Spike and slab,

$$p(e_i) = \alpha\mathcal{N}(0, v_1) + (1-\alpha)\mathcal{N}(0, v_2),$$

- Student-t $p(e_i) = \mathcal{S}t(\nu, 0, \sigma)$,







Spike&Slab

Spike and slab prior of the noise implies:

$$p(e_i, l_i) = \mathcal{N}(0, v_1)^{l_{1,i}} \mathcal{N}(0, v_2)^{l_{2,i}},$$
$$p(y_i, l_i | x_i) = \mathcal{N}(ax_i + b, v_1)^{l_{1,i}} \mathcal{N}(ax_i + b, v_2)^{l_{2,i}},$$

Do the point outside of the line follow the $ax_i + b$ curve?

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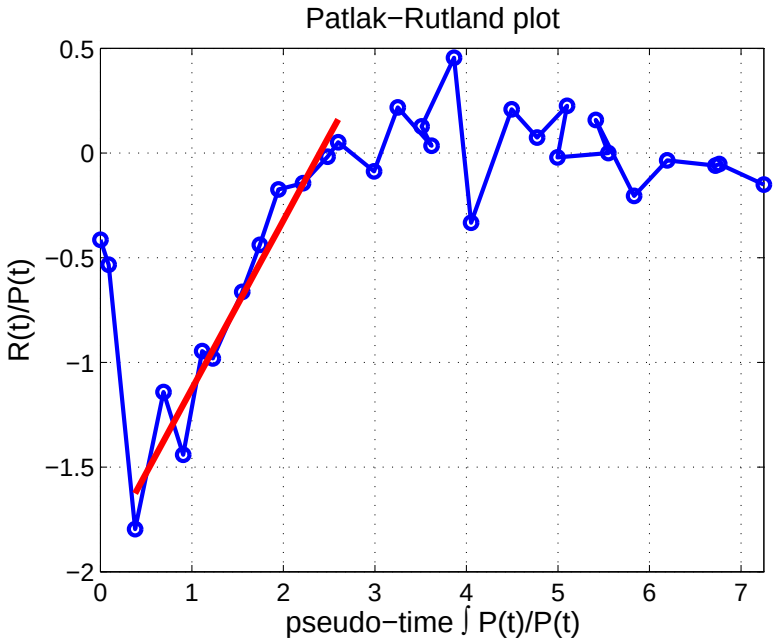
Do the point outside of the line follow the $ax_i + b$ curve?

What model do they follow?

$$p(y_i, l_i | x_i) = \mathcal{N}(ax_i + b, v_1)^{l_{1,i}} \, p(y_{\text{nom}} | l_i = \epsilon_2)^{l_{2,i}},$$

Possibilities:

$$p(x_i | l_i = \epsilon_2) = \mathcal{U}(y_{min}, y_{max}),$$
$$p(x_i | l_i = \epsilon_2) = \mathcal{N}(a_2 x, v(x_i, \psi)),$$
$$\vdots$$



Be creative.

Assignment

Load data Patlak.mat
35 studies with:

xpr	x axis
ypr	y axis
name	name of the study
int_start	ignore
int_end	ignore

Assignment	points
outlier model	5
alternative model	5